

What's My Age Again? Exploring the Impact of Age on the Enfacement of Current State-of-the-Art Avatars

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ABSTRACT

In this work, we examine how study participants experience enfacement toward high fidelity state-of-the-art avatars of different ages while assessing perception of avatar appearance. While most traditional literature focuses on the feeling of ownership and agency over an avatar's body, recent developments in real-time facial tracking capabilities have brought about budding interest in the feeling of ownership, agency, and self-identification with an avatar's face. We present a study in which participants viewed avatars of different ages – young, middle-aged, and older – on a desktop monitor fitted with advanced real-time tracking and rated the degree of enfacement they felt toward each avatar as well as the uncanniness of the avatar's appearance.

Keywords: Embodiment, Enfacement, Avatars, Uncanny valley.

1 INTRODUCTION

For decades animated avatars have been used as a mechanism through which humans and agents are represented in virtual spaces. When it comes to designing a uniquely immersive environment, users must experience embodiment while controlling a self-avatar. One component of embodying an avatar is called the *enfacement illusion*, or having the experience of ownership of and agency over a virtual avatar's face. Albeit, little study has been conducted on how people experience enfacement toward virtual avatars. We focus on pushing toward a deeper understanding of the enfacement illusion and how it may potentially affect one's behavior in a virtual environment.

In this study we focus on the enfacement of animated self-avatars displayed via a desktop monitor. As video conferencing applications have gained in popularity, self-avatars have been presented as an alternative to the "camera-off" option provided by these platforms [5]. Zoom fatigue is correlated with viewing one's own appearance on screen for an extended period of time [8], and avatar-based systems can potentially provide relief while ensuring one's privacy and improving subjective engagement via a sense of presence, self-identification, and body ownership.

Tsakiris [9] was the first to explore the enfacement illusion in cognitive science, showing that one can view another image as the self in the presence of synchronous multisensory stimuli. A few studies conducted on avatars on desktop screens and in virtual reality (VR) have shown enfacement can occur with avatars [2], regardless of avatar stylization or self-similarity [7]. However, this is the extent of the literature, and there are many complexities that arise from the plasticity of avatars that have the potential to affect enfacement.

One such complexity we explore is the possibility of an age mismatch between the user and an embodied avatar, which leads to our

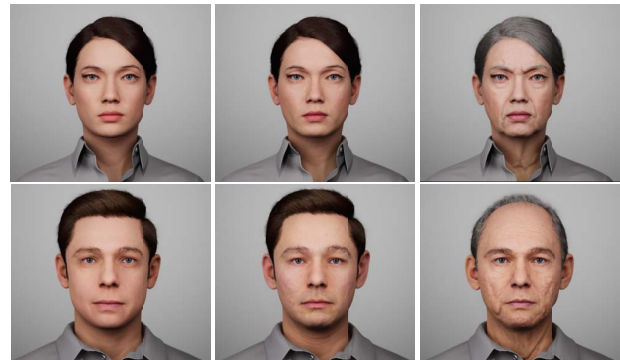


Figure 1: The MetaHuman avatars used in our experiment. Top left: young female; top middle: middle-aged female; top right: older female; bottom left: young male; bottom middle: middle-aged male; bottom right: older male.

research question: does the age of a given virtual self-representation affect the enfacement illusion? Here, we examine how users experience the enfacement of avatars of different ages – young (20s), middle aged (40s), and older (60s+) – while viewing them on a desktop monitor as if looking in a mirror. While it has already been demonstrated that embodiment occurs regardless of the age of a given avatar [1], this result has not been considered in the context of self-identification with the face of a highly realistic avatar fitted with real-time facial tracking. Furthermore, an avatar's appearance can positively or negatively affect embodiment [3], so we also consider how the uncanny valley effect may affect enfacement in this context. To our knowledge, there is no precise knowledge of how this effect plays a role in both embodiment and enfacement.

2 EXPERIMENT

We chose a within subjects design based on Tsakiris's traditional methodology from cognitive science [9], which is also followed by more recent work on the enfacement of virtual characters [7]. Before beginning the study, participants signed a consent form and answered a short pre-test questionnaire so that we could ascertain their demographic information. Participants answered "to what gender do you identify," and the experimenter selected the appropriate avatar gender to represent the participant. Figure 1 shows the avatars used in this study. Participants were then seated in front of a computer screen and told that a virtual agent would guide them through the experiment. The virtual agent was not pictured on the screen, but communicated via speakers so participants could focus on their virtual appearance without distraction.

When the experiment began, the virtual avatar meant to represent the user appeared on the screen. Real-time facial tracking was used to capture facial expressions and head movements from the users. Participants were asked to perform three different tasks during the study. They were to: (1) trace shapes that appeared on the screen with their nose, (2) answer questions the agent asked, and (3) repeat

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tongue twisters. These tasks were designed to elicit a range of head movements and facial expressions from users in order to enhance the sensation of agency over the virtual avatar.

After the conversation between the user and the agent, a simulation of a boulder swinging and hitting the virtual avatar in the head occurred. We expected that, if participants felt embodied, they would respond as if they themselves were being hit in the head by the swinging boulder. Once this occurred, the program ended and a 35-item questionnaire was given to participants to assess feelings of embodiment and the uncanny valley effect. Since our study was designed to be within subjects, participants experienced 3 different counterbalanced conditions on the age of the avatar and filled out the same post-test questionnaire after each condition. The experiment took approximately 45 minutes.

Twenty eight volunteers (20 male, 8 female) aged 18–64 (31.7 ± 12.3) participated in our study, recruited via an institutional-wide email advertisement. Participants gave informed, written consent and received a €5 book voucher for participating.

2.1 Questionnaires

We used an adapted version of Gonzalez-Franco and Peck's embodiment questionnaire [3] to measure the level of enfacement users felt toward their avatar. It contained 16 items aimed at assessing the subcomponents of embodiment that could apply to the feeling of enfacement: body ownership (items 1–3), agency (items 4–7), tactile sensation (item 8), body location (items 9–11), external appearance (items 12–13), and response to external stimuli (items 14–16). Each item was scored on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Ho and MacDorman's alternative Godspeed indices [6] were used to measure the uncanny valley effect. It contained 19 items that were aimed to assess the factors that contribute to the uncanny valley effect: perceived humanness (items 1–6), eeriness (items 7–14), and attractiveness (items 15–19). Each item was scored on a five-point scale with semantic differential items on the opposite end of the scale.

2.2 Results

The data from our enfacement questionnaire consists of scores for each component of enfacement (body ownership, agency, tactile sensation, location, appearance, and threat response) and one total overall enfacement score. We ran Shapiro-Wilk Tests to determine normality and ran Friedman Tests where normality was violated.

We ran a repeated measures analysis of variance with within subjects factors of condition (age: young, middle-aged, and older), age grouping (younger than 30, 30 and older), technical familiarity, and game usage over the **total enfacement** score. No significant effects were found, $F(2, 18) = 0.18, p = 0.84, \eta_p^2 = 0.02$. Mean total enfacement scores and standard deviations were $M = -0.13, SD = 0.39$ for the young, $M = -0.07, SD = 0.45$ for the middle-aged, and $M = -0.11, SD = 0.44$ for the older avatar.

No significant difference was found between the individual ratings of **body ownership, agency, tactile sensation, self-location, appearance, or threat response** between the different aged avatars. Thus, we performed no further analyses.

To analyze the data from the uncanny valley questionnaire, we first ran Shapiro-Wilk Tests to determine normality. These tests revealed our data not to be normal, thus we chose to run Friedman Tests for each component – human-likeness, eeriness, and attractiveness – to determine if age affected the uncanny valley scores.

No significant difference was found between ratings for **human-likeness**, with mean scores of: $M = 3.09, SD = 0.92$ for the young, $M = 3.22, SD = 0.69$ for the middle-aged, and $M = 3.23, SD = 0.69$ for the older avatar. Additionally, no significant difference was found between ratings of **eeriness** with mean scores of: $M = 2.63, SD = 0.86$ for the young, $M = 2.82, SD = 0.63$ for the middle-aged, and $M = 2.75, SD = 0.63$ for the older avatar.

There was a statistically significant difference in perceived attractiveness of the avatars depending on the age, $X^2(2) = 15.89, p < 0.01, W = 0.29$, with mean scores of: $M = 3.64, SD = 0.59$ for the young, $M = 3.46, SD = 0.67$ for the middle-aged, and $M = 2.94, SD = 0.5$ for the older avatar. Post hoc analyses with Wilcoxon signed-rank tests were conducted with a Bonferroni correction applied, resulting in a significance level at $p < 0.017$. There was a statistically significant difference in the attractiveness ratings between the young and old avatar ($Z = -3.73, p < 0.01$) and the middle-aged and old avatar ($Z = -3.09, p < 0.01$), but no difference between the young and middle-aged avatars.

3 DISCUSSION AND CONCLUSIONS

In this work, we explored whether study participants would experience enfacement toward screen-based avatars of different ages and whether it was affected by the uncanny valley effect. Results from our adapted enfacement questionnaire revealed that enfacement was not particularly strong under our experimental conditions as the mean total enfacement score across all conditions was -0.10 , showing that participants did not strongly self-identify with nor did they not self-identify with their given self-avatars.

Interestingly, participants felt a similar sense of enfacement regardless of the given self-avatar's age. This finding is consistent with Banakou et al.'s [1] work, in which participants experienced similar levels of embodiment toward self-avatars regardless of age. Both Gonzalez-Franco et al. [4] and Higgins et al. [5] have shown enfacement to occur regardless of self-similarity as well. Additionally, participants did not perceive differences in human-likeness and eeriness between the different aged avatars.

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